Travis Axelrod (00:00):

Good afternoon everyone, and welcome to Tesla's First Quarter 2025 Q&A Webcast. My name is Travis Axelrod, Head of Investor Relations, and I'm joined today by Elon Musk, Vaibhav Taneja, and a number of other executives. Our Q1 results were announced at about 3PM Central Time in the update deck we published at the same link as this webcast.

(00:19)
During this call, we will discuss our business outlook and make forward-looking statements. These comments are based on our predictions and expectations as of today. Actual events or results could differ materially due to a number of risks and uncertainties, including those mentioned in our most recent filings with the SEC.

(00:37)
During the question-and-answer portion of today's call, please limit yourself to one question and one follow-up. Please use the raise hand button to join the question queue. Before we jump into Q&A, Elon will be providing an update. Elon?

Elon Musk (00:50):

Hello, everyone. Well, there's never a dull moment these days. Thanks for sure. Every day is going to be exciting.

(00:59)
As some people know, there's been some blowback for the time that I've been spending in government with the Department of Government Efficiency or DOGE. I think the work that we're doing there is actually very important for trying to sprain in the insane deficit that is leading our country, the United States, to destruction. And the DOGE team has made a lot of progress in addressing waste and fraud.

(01:22)
The natural blowback from that is those who were receiving the wasteful dollars and the fortunate dollars will try to attack me and DOGE team and anything associated with me. But then I'm really left with two choices. Should we just let the waste and fraud continue? And it was continuing to grow at a really unsustainable pace that was bankrupting the country. Or to fight the waste and fraud and try to get the country back on the right track? And I believe the right thing to do is to just fight the waste and fraud and get the country back on the right track and working together with President Trump and his administration. Because if the ship of America goes down, we all go down with it, including Tesla and everyone else. So, I think this is critical work.

(02:20)
Now, the protests that you'll see out there, they're very organized, they're paid for. They're obviously not going to say, admit that the reason that they're protesting is because they're receiving fraudulent money or that they're the recipients of wasteful

largesse, but they're going to come up with some other reason. But the real reason for the protests, the actual reason is that those receiving the waste and fraud wish to continue receiving it. That is the real thing that's going on here, obviously.

(03:04)
Now that said, I do think there's the large slug of work necessary to get the DOGE team in place and working in the government to get the financial house in order is mostly done. And I think starting probably next month, May, my time allocation to DOGE will drop significantly. I'll have to continue doing it for, I think, probably the remainder of the President's term, just to make sure that the waste and fraud that we stop does not come roaring back, which will do if it has the chance.

(04:02)
So, I think I'll continue to spend a day or two per week on government matters for as long as the President would like me to do so and as long as it is useful. But starting next month, I'll be allocating probably more of my time to Tesla and now that the major work of establishing the Department of Government Efficiency is done.

(04:36)
So, at Tesla, we've gone through many, many crises over the years and actually been through many near-death experiences. We were probably on the ragged edge of death at least on maybe a dozen times. It's been so many times. This is not one of those times. We're not on the ragged edge of death, not even close. But there are some challenges, and I expect that this year will be, there will probably be some unexpected bumps this year. But I remain extremely optimistic about the future of the company.

(05:26)
The future of the company is fundamentally based on large-scale autonomous cars and large-scale, large volume, vast numbers of autonomous humanoid robots. So the value of a company that makes truly useful autonomous humanoid robots and autonomous useful vehicles at scale at low cost, which is what Tesla is going to do, is staggering. I continue to believe that Tesla, with excellent execution, will be the most valuable company in the world by far. But that's an important if, we must execute well. But if we do execute well, I think Tesla will be the most valuable company in the world by far. It may be as valuable as the next five companies combined.

(06:39)
But there'll be a few bumps along the road before that happens. I said I think on the last earnings call that we'll start to see the prosperity of autonomy take effect in a material way around the middle of next year. We expect to have these … be selling fully autonomous rides in June in Austin, as we've been saying for now several months. So that's continued, but the real question from financial standpoint is when does it really become material and affect bottom line of the company and start to be a fundamental part of the … When does it move the financial needle in a significant way? That's probably around the middle of next year, second half of next year. And then once it

does start to move the financial needle in a significant way, it will really go exponential from there.

(07:45)
So that's, I'd encourage people to look beyond some sort of bumps and bundles of each road immediately ahead of us. Lift your gaze to the bright shining sort of down the hill, I don't know, some Reagan-esque imagery. And that's where we're headed and not too distant future like I said. Kind of next year or two. So, let's see.

(08:35)
With respect to supply chain risk, something that Tesla has been working on for several years is to localize supply chains. This actually makes sense from a cost standpoint and from a logistics risk standpoint, is to have the supply chains be at least located on the continent in which the car is built. So we are, I think the least company, the least affected car company with respect to tariffs, at least in most respects. I mean, it remains to be seen. Now, tariffs are still tough on a company when margins are still low. But we do have localized supply chains in North America, Europe, and China. So that puts us in a stronger position than any of our competitors.

(09:50)
And undoubtedly, I'm going to get a lot of questions about tariffs, and I just want to emphasize that the tariff decision is entirely up to the President of the United States. I will weigh in with my advice with the President, which he will listen to my advice, but then it's up to him, of course, to make his decision. I've been on the record many times saying that I believe lower tariffs are generally a good idea for prosperity, but this decision is fundamentally up to the elected representative of the people being the President of the United States. So I'll continue to advocate for lower tariffs rather than higher tariffs, but that's all I can do.

(10:38)
Now let me walk you through why I'm so excited about the future of Tesla. So, first of all, autonomy. The team and I are laser focused on bringing robotaxi to Austin in June. Unsupervised autonomy will first be solved for the Model Y in Austin. And then … Actually you should parse out the terms robotic taxi or robotaxi and just generally like what's the Cybercab because we've got a product called the Cybercab and then any Tesla which could be an S3 extra wide that is autonomous is a robotic taxi or robotaxi. It's very confusing. So the vast majority of the Tesla fleet that we've made is capable of being a robotaxi or robotic taxi.

(11:39)
Once we can make the whole system work where you can have paid rides fully autonomously with no one in the car in one city, that is a very scalable thing for us to go broadly within whatever jurisdiction allows us to operate. So, because what we're solving for is a general solution to autonomy, not a city specific solution for autonomy, once we make it work in a few cities, we can basically make it work in all cities in that legal jurisdiction. Once we can make it based to work in a few cities in America, we can

make it work anywhere in America. Once we can make it work in a few cities in China, we can make it work anywhere in China, likewise in Europe, limited only by regulatory approvals.

(12:34)
So, this is the advantage of having a generalized solution using artificial intelligence and an AI chip that Tesla designed specifically for this purpose as opposed to very expensive sensors and high precision maps of a particular neighborhood where that neighborhood may change or often changes and then the car stops working. So, we have a general solution instead of a specific solution.

(13:09)
Then, with regards to Optimus, we're making good progress in Optimus. We expect to have thousands of Optimus robots working in Tesla factories by the end of this year, 10 years forward. And we expect to scale Optimus up faster than any product, I think, in history, to get to millions of units per year as soon as possible. I think I feel confident in getting to a million units per year in less than five years, maybe four years. So, by 2030, I feel confident in predicting a million Optimus units per year. It might be 2029.

(14:14)
Let's see with respect to energy, energy business is doing very well. The Megapack enables utility companies to output far more total energy than would otherwise be the case. When you think of the energy capability of a grid, it's much more than, say, total energy output per year. If the powerplants could operate at peak power for all 24 hours, as opposed to being at half power, sometimes a quarter power at night, then you could double the energy output of existing power plants. But in order to do that, you need to buffer the energy, so that you can charge up something like a battery pack at night and then discharge into the grid during the day. So, this is a massive unlock on total energy output of any given grid over the course of a year. And utility companies are beginning to realize this and are buying in our Megapacks at scale.

(15:31)
So, at this point, a gigawatt class battery is quite a common thing. We have many orders and offer for gigawatt and beyond batteries. And we expect the stationary energy storage business to scale ultimately to terawatts per year. So very, very good numbers. Now, Q1, first quarters of a year are usually pretty tricky. Because it's usually the worst quarter of the year because people don't want to go buy a car in the middle of winter during a blizzard. So we picked Q1 as a good quarter to do a cutover to the new version of the Model Y and we changed production of the world's bestselling cars with … Remember the Model Y is the bestselling car of any kind on earth with a 1.1 billion unit per year output of a single model. And we did this changeover at the same time in factories all across the world. So congratulations to the Tesla team on an amazing job in pulling off what is a very difficult transition. So yeah, it's really very impressive work. So, yeah.

(17:10)
In conclusion, while there are many near-term headwinds for us and the border industry, the future for Tesla is brighter than ever. The value of the company is delivering sustainable abundance with our affordable AI powered robots. I like this phrase, sustainable abundance for all. If you say, like, what's the ideal future that you can imagine? That's what you'd want. You'd want abundance for all in a way that's sustainable. It's good for the environment. Basically, this is the happy future. If you say what's the happiest future you can imagine. One which is that would be a future where there's sustainable abundance for all. Closest thing to heaven we can get on Earth, basically.

(18:06)
So, thank you again to the Tesla team for all their efforts of the challenging time. I look forward to continuing to lead the team to great success in the future.

Travis Axelrod (18:22):

Great. Thank you very much, Elon. Before we move on, Vaibhav has some opening remarks as well.

Vaibhav Taneja (18:28):

Thanks, Elon. As Elon mentioned, in Q1, we achieved something which has never been undertaken in the automotive industry of updating all our factories for the bestselling car in the world, all at the same time. And this is, people don't understand, this was not a small feat. We're not aware of anybody else being able to do the bestselling car all at once within a quarter. And that too hitting all the timelines which we had established at the beginning. So, big kudos to the team for making this happen. Additionally, we also hit record gross profit for energy storage business in the quarter.

(19:09)
Now, getting back into the business, there has been a lot of speculation as to the reasons for decline of our vehicle deliveries in the first quarter. We had previously guided that we will be updating all factories and this will lead to several weeks of lost production, which did happen as planned. The ripple effect of the change is not having enough new Model Y available in most markets for people to see and experience till the last few weeks of the quarter. Additionally, the negative impact of vandalism and unwanted hostility towards our brand and our people had an impact in certain markets.

(19:48)
Despite this, we were able to sell out legacy Model Y in US, China, and a few other markets within the world. And again, just so that people understand, we were producing the legacy Model Y till middle to end of February. And we switched over and we were able to still sell out within that period. So, again, big achievement by all the people at Tesla to make it happen.

(20:18)
We have a very extremely competitive vehicle lineup, which with most vehicles going through a recent update, and add to that, if it wants an FSD, you have a personal chauffeur which can take you almost anywhere under supervision. There are numerous stories shared by customers ranging from how it has improved their daily commute, to providing mobility to customers with disabilities, to giving older customers the ability to travel comfortably and independently. Not only is FSD Supervised safer than a human driver, but it is also improving the lives of individuals who experience it. And again, this is something you have to experience, and anybody who has experience just knows it. And we've been doing a lot lately to try and get those stories out, at least on X, so that people can see how other people have benefited from this.

(21:17)
Now, coming into some of the financial stuff, auto margins declined sequentially primarily due to a reduction in the total number of deliveries, lower fixed cost absorption due to factory change awards and lower regulatory credit revenues offset by a slight increase in pricing due to the launch of new Model Y despite incentives which we had to sell legacy Model Y.

(21:46)
Our energy storage business, like I said before, has achieved yet another milestone of create highest gross profit in the quarter. This was despite sequential decline in deployments.

Vaibhav Taneja (22:00):

The importance of this business, as Elon mentioned, is pretty profound, especially in this environment, because in order for grids to work properly with the demands from AI and all this, you need some more stability. And this is, by far, the simplest and best solution, which we are aware of, which can help do this. And we've also developed certain unique solutions to help our customers to achieve this. Additionally, on the Powerwall side, we've been selling the new Powerwall 3 and it's been received with very good reception from customers and to the extent that we are currently supply constrained.

(22:47)
On services and other margins, they were slightly down sequentially, primarily because of the pressure on the used card business and insurance business. Note that we continued our journey to improve profitability in our services and [inaudible 00:23:01] business through better level of productivity. As previously discussed, our operating expenses continued to increase sequentially, primarily due to our AI-related initiatives, including Optimus, and also cost of development for our vehicle programs including Cybercabs, semi, and cheaper models. These expenses flow through R&D. We believe, even in the current environment, it is the right strategy making investments in these areas to position us for the long term. These increases were offset by decreases and the CNA from changes in our molecular flow program.

(23:41)
Other income reduced significantly on a sequential basis. The primary reason was … well, Bitcoin mark-to-market loss in Q1 versus gain in Q4 resulting in a 472 million drop. The remainder of the change is because of FX [inaudible 00:24:00]. With the adoption of the new mark-to-market standard for Bitcoin, we expect increased volatility in our other income in addition to the FX [inaudible 00:24:10]. I know tariffs is the hottest topic, which people talk about, and it has various impacts to our business. As Elon mentioned, on the wake business, we've been on this journey of regionalization for years. Specifically in the US, Model Y has been rated the most American made car on cars.com made in America index three years in a row. This, in part, is of all the work which team has been doing over the years, and to the extent that today, if you look at our vehicle lineup in US, where about approximately, on a weighted average basis, 85% USMCA compliant.

(24:57)
So, like Elon said, this definitely gives us a bigger edge as compared to our other OEMs in terms of managing the tariffs, but we're not immune, because when the Section 232 auto tariffs become effective in May, which includes Canada and Mexico, and Canada and Mexico has been part of our regionalization [inaudible 00:25:19], they will have an impact on profitability. And I know research modeling on this impact has been about a couple of thousand give or [inaudible 00:25:27], which is pretty much in line with what we've been forecasting. The impact of tariffs on the energy business will be outsized since we source LFP battery cells from China. We are in the process of commissioning equipment for the local manufacturing of LFP battery cells in the US, however, the equipment which we have can only service a fraction of our total installed capacity [inaudible 00:25:56].

(25:56)
We've also been working on securing additional supply chain from non- China-based suppliers, but it will take time. Also note that [inaudible 00:26:07] irrespective of all the impact on US from tariffs on the energy business, we do have mega-factory China, which just started operations in Q1, and that should take care of our business outside of the US. There's also an important impact of tariffs on our capital investments. I know this is going to sound counterintuitive since in order to [inaudible 00:26:35] manufacturing or expand lines, we have to bring equipment from outside the US, because there is not that much capacity in US. And current trade environment, such equipment being brought in is subject to-

Elon Musk (26:48):

The expense is bringing it in from China right now.

Vaibhav Taneja (26:49):

Exactly. And the reality is that China has the basic one which has the most capacity to provide this [inaudible 00:27:01]. Our CapEx [inaudible 00:27:05], inclusive of modern tariffs, even with the optimization we have tried to do, it is forecasted to be still in excess of 10 billion this year. We are still evaluating what more to do on this one. To summarize, we have near-term challenges in our business due to tariffs and brand image. We think our strategy of providing the best product at a competitive price is going to be a winner, and this is the reason we're still focused on bringing cheaper models to market soon, which start of production still planned for June. Additionally, the advancement in FSD related features, including pilot Robotaxi launch in Austin later this year, should help create a new area of demand. I would like to thank everyone at Tesla and our customers.

Travis Axelrod (27:54):

Fantastic, thank you very much [inaudible 00:27:56]. Now we will move on investor questions. We will start with questions from say.com. The first question is, what are the highest risk items on the critical path to Robotaxi launch and scaling?

Joshua (28:14):

This is Ashok?

Travis Axelrod (28:16):

Yeah, we've got Ashok on the line.

Elon Musk (28:18):

Sure. Well, just … we must just [inaudible 00:28:23] by the … disambiguate the Cybercab from Robotaxi once again. Because the Teslas that will be fully autonomous in June, in Austin, probably Model Ys, so that's currently on track to be able to do paid rides, fully autonomously, in Austin in June, and then to be in any other cities in the US by the end of this year. It's difficult to predict the exact ramp week by week or month by month, except that it will ramp up very quickly. So it's going to be like basically an S-curve where it's very difficult to predict the intermediate slope of the S-curve, but you kind of know where the S-curve is going end up, which is the vast majority of the Tesla fleet being autonomous.

(29:35)
So that's why I feel confident in predicting large-scale autonomy around the middle of next year, certainly the second half of next year, meaning I predict that there'll be millions of Teslas operating fully autonomously in the second half of next year. It does seem increasingly likely that there will be a localized parameter set perhaps sort of … especially for places that have, say, very snowy weather. Like say if you're in the Northeast or something, like there's … you can think of it, it's kind of a human. You could be a very good driver in California, but are you going to be also a good driver in a blizzard in Manhattan? You're not going to be as good.

(30:52)
So there is actually some value in … you can still drive, but your probability of an accident is higher. So it is increasingly obvious that there's some value to having a localized set of parameters for different regions and localities. But the output [inaudible 00:31:23] the [inaudible 00:31:23] category, it's not the required category. Again, really the car is just very much like the human. It's digital neural nets and cameras, and humans operate with biological neural nets and eyes. And so the same strengths and weaknesses will be present. Or a digital neural net and cameras versus a biological neural net and eyes. Ashok, if you'd like to elaborate on that.

Joshua (31:56):

Yeah. Speaking to the location-specific models, we still have a generalized approach and you can see that from our deployment of FSD supervised in China, where with this very minimal data that's channel-specific, the models generalize quite well to completely different driving styles. That just shows that the AI-based solution that we have is the right one, because if you had gone down the previous rule-based solutions or more hard-coded HD map-based solutions, it'll have taken many, many years to get China to work. You can see those in the videos that people post online themselves. So the generalized solution that we are pursuing is the right one that's going to scale well. And you can think of this location-specific parameters that you don't [inaudible 00:32:43] to as a mixture of experts, and if you are sort of familiar with the AI models, Grok and others, they all use this mixture of experts to sort of specialize the parameters to specific tasks while still being general.

(32:57)
This makes the model use [inaudible 00:33:02] amount of compute to solve for the diversity of tasks that it has to solve. In terms of addressing the question that asked for what are the critical things that need to get right? One thing I would like to note is validation. Self-driving is a long-tail problem where there can be a lot of edge cases that only happen very, very rarely. Currently, we are driving around in Austin using our QA fleet, but then it's super rare to get interventions that are critical for Robotaxi operation. And so, you can go many days without getting any single intervention, so you can't easily know whether you're improving or regressing in your capacity. And we need to build out sophisticated simulations, including neural network-based video generation. That's all happening in the background to make sure that we deliver a safe product and we are able to measure our safety even though we can't just exceed [inaudible 00:33:55] driving around the block or something like that.

Elon Musk (33:59):

I mean very basic terms. If we're sitting at an accident every 10,000 miles, well then you have to drive 10,000 miles, on average, before you get an accident or an intervention. So it's like, "Okay." We must be really … people must be very wigged out by the sheer number of Teslas doing circuits in Austin right now. We're like … it's going to look pretty bizarre.

Joshua (34:29):

[inaudible 00:34:28] Some people are chasing us right now.

Elon Musk (34:32):

Yeah, there's just always a convoy of Teslas going, just going all over to Austin in circles. But I just can't emphasize this enough. In order to figure out long tail things, if it's one in 10,000, let's say it's one in 20,000 miles, or one in 30, the average person drives 10,000 miles in a year. So now try to compress that test cycle into a matter of a few months, that means you need a lot of cars during a lot of driving in order to compress that, or just do in a matter of a month what would normally take someone a year.

Vaibhav Taneja (35:16):

I would just also add that if you haven't looked at those videos coming out of China, people are [inaudible 00:35:24]-

Elon Musk (35:24):

Oh, yeah, those videos are amazing.

Vaibhav Taneja (35:25):

Yeah, they're putting it to real test. I mean, they're dark [inaudible 00:35:30]-

Elon Musk (35:30):

Frankly, I think the Chinese consumer might be the most demanding consumer, and actually the customers in China are awesome. They have a lot of fun with the cars. I saw one guy take a Tesla, autonomous, on a narrow dirt road across a mountain. And I'm like, "[inaudible 00:35:54] a very brave person." And the Tesla's driving along on a road with no barriers where, if it makes a mistake, he's going to plunge to his doom. But it worked.

Travis Axelrod (36:06):

Great. Thank you.

Speaker 1 (36:07):

And if the question was on CyberCab itself, we're in B sample validation now.

Elon Musk (36:14):

Yeah, yeah. We should ask that question too.

Speaker 1 (36:15):

Yeah, we have our first big builds coming at the end of this quarter within Q2, and then, in the coming months, we start to large scale installation of all the equipment in Giga Texas with still on schedule for production next year.

Vaibhav Taneja (36:34):

And I just want to also clarify, because I think people don't understand the thing that there is no new building being built, and where is Cybercab going to [inaudible 00:36:42]-

Elon Musk (36:42):

It's in the Giga [inaudible 00:36:43].

Vaibhav Taneja (36:43):

It's literally same factory.

Elon Musk (36:43):

Yeah. Yeah.

Speaker 1 (36:48):

It's happening and people don't know. It's just happening upstairs and along the long lines while we're still building the Model Ys and Cybertrucks every day.

Elon Musk (36:54):

Yeah. It's worth noting that the Tesla Gigafactory in Austin is three times the size of the Pentagon.

Speaker 1 (37:02):

Including the garden.

Elon Musk (37:04):

Yeah, including the ground zero garden. So because of the Pentagon, like this building used to look big. Not anymore.

Travis Axelrod (37:16):

Great, thank you very much. The next question is when will FSD Unsupervised be available for personal use on personally owned cars.

Elon Musk (37:30):

But before the end of this year. Not necessarily … let's say, within the US … we do want to test … at Tesla, we're absolutely hardcore about safety. We go to great lengths to make the safest car in the world, and have the lowest accidents per mile, and fewest lives lost, so we want to be very careful. And we want autonomy to be definitively safer than manual driving. So it's not enough that it'd just be as safe, it needs to be meaningfully safer than if it's cars manually driven. And we want to confirm that there's not something … we just want to be cautious with the roll-out. We don't want to jump in at the deep end with an army. But that said, I think we should be able to have it work in several studies, later this year, for personal use. The acid test being you should be able to … can you go to sleep in your car and wake [inaudible 00:38:46] your destination? And I am confident that will be available in many cities in the US by the end of this year.

Travis Axelrod (38:57):

Great, thank you very much. The next question is, is Tesla still on track for releasing more affordable models this year or will you be focusing on simplified versions to enhance affordability, similar to the rear wheel drive Cybertruck?

Lars (39:11):

Yeah, we're still planning to release models this year. As with all launches, we're working through the last minute issues that pop up. We're knocking them down one by one. At this point, I would say that ramp maybe might be a little slower than me at home initially, but there's nothing … just given that turmoil that exists in the industry right now, but there's nothing that's blocking us from starting production within the timeline laid out in the opening remarks.

(39:35)
And I will say it's important to emphasize that, as we've said all along, the [inaudible 00:39:40] of our factories is the primary goal for these new products. And so flexibility of what we can do within the form factor and the design of it is really limited to what we can do on our existing lines rather than building new ones. But we've been targeting the low cost of ownership. Monthly payment is the biggest differentiator for our vehicles, and that's why we're focused on bringing these new models with the lowest price to the market within the constraints I just highlighted.

Travis Axelrod (40:07):

Great, thank you very much. The next question is, does Tesla see Robotaxi as a winner-take-most market? And as you approach the Austin launch, how do you expect to compare against Waymo's offering, especially regarding pricing, geofencing and regulatory flexibility?

Elon Musk (40:23):

Well, okay. The issue with Waymo's cars is it costs way more money. [inaudible 00:40:38] But that is the issue. The cars are very expensive, made in low volume. Teslas are, I don't know, probably cost a quarter, 20% [inaudible 00:40:54] a quarter what a Waymo costs, and made in very high volume. So, ironically, we are the ones that made the bet that a pure AI solution with cameras, and what do you have, the car actually will listen for sirens and that kind thing, is the right move. And Waymo decided that an expensive sensor suite is the way to go even though Google's very good at AI. It's ironic.

(41:23)
And it is worth noting that Tesla's built an incredible AI software team and AI hardware chip design team. Prospect nothing. We didn't acquire anyone. We just built it. So yeah, it's really … I mean, I don't see anyone being able to compete with Tesla at present. I'm sure that'll change eventually, but at least as far as I'm aware, Tesla will have, I don't know, 99% market share or something ridiculous. That ninety-something percent, at least, I don't know, something might change, but if we have millions of cars deployed next year, and unless others have millions of cars deployed … unless we're blocked by regulatory situations, it won't be long. I mean, in a few years we'll have 10 million autonomous cars on the roads and counting.

Vaibhav Taneja (42:41):

And the other thing which people forget is we're not just developing the software solution, we are also manufacturing the cars.

Elon Musk (42:48):

Yeah.

Vaibhav Taneja (42:48):

And like you know what Waymo has, they're taking cars, then trying to put [inaudible 00:42:54]-

Elon Musk (42:53):

Way more money.

Vaibhav Taneja (42:56):

We don't do that. So that definitely gives us a big, big up. And like Elon said, we already have a big existing fleet, which hopefully with the software update, could become autonomous.

Elon Musk (43:09):

With the software update, it will become autonomous. To be clear, the Model Ys that we're talking about being autonomous in Austin in June are the Model Ys we make currently. There's no change to it.

Vaibhav Taneja (43:23):

I think people don't appreciate that the car, which they can buy today-

Elon Musk (43:26):

The car that they have.

Vaibhav Taneja (43:27):

… or the car they have is capable of these kind of things.

Joshua (43:33):

In fact, it does drive autonomously from the factory to the end of line. Every car, nowadays.

Elon Musk (43:38):

Yeah.

Speaker 1 (43:39):

[inaudible 00:43:39] through the tunnel, the Model Ys. Everything.

Elon Musk (43:41):

Right. Exactly. It is being put to … it's doing useful work, fully autonomously, at the factories. As Ashok was mentioning, the cars driving themselves from end of line to where they're supposed to be picked up by a truck to be taken to a customer.

Elon Musk (44:00):

… Customer, and I'm confident also that, later this year, the first Model Y will drive itself all the way to the customer. From our factory in Boston and our one in… Here in

Fremont, California, I'm confident that, from both factories, we'll be able to drive directly to a customer from a factory.

Lars (44:27):

Cool delivery.

Elon Musk (44:28):

Yeah. Literally goes from the end of line and drives itself to your house.

Speaker 2 (44:33):

It's important to note, in the factories, we don't have dedicated lanes or anything. People are coming in and out every day, trucks delivering supplies, parts, construction. For the-

Elon Musk (44:42):

And people can film it. By the way, you can see this from the road. It's not covered.

Speaker 2 (44:45):

Exactly.

Elon Musk (44:47):

In these videos, people will take videos online and anyone who wants to go see it can just drive past our Fremont factory and see the autonomous cars-

Speaker 2 (44:54):

[inaudible 00:44:55].

Elon Musk (44:54):

… driving themselves. Yeah, and they drive themselves and they put themselves in the exact right spot to be picked up.

Speaker 2 (44:59):

Yeah, the logistics yard is right there in the open. We don't move it again to another lane.

Elon Musk (45:11):

But they go to a specific spot. Find a spot.

Speaker 2 (45:11):

Yeah.

Elon Musk (45:11):

Yeah, that's just a routine everyday thing now.

Travis Axelrod (45:16):

Great, thank you very much. The next question is can you please provide an update on the unboxed method and how that is progressing?

Lars (45:23):

Sure. It's progressing, absolutely. As I mentioned just a minute ago, it is the basis for our Cybercat manufacturing process. It's really what we changed in order to allow the low cost of production and also get the super high levels of automation. Really, levels of automation that are unheard of in the vehicle manufacturing scale. This is not something that, when you see it be produced, you'll think of in terms of like, wow, that's how the car has been built for 100 years. It's really something we've changed.

(45:51)
In the past year, we've been focusing on a lot of key development areas, like marrying these large sub-assemblies together in a precise way, in an accurate way. We've also de-risked things like corrosion of uncoated aluminum structures, the ceiling across the seams of the vehicle, and when you marry the assembled components. And we've even done early crash testing and we've proven that it's going to be just as safe as every other car we build.

(46:16)
With all that combined, we go into the builds that we have at the end of this quarter for the CyberCat product, and that's the next real big test of full scale integration of the unboxed process. And that's where we are, so you'll see them in testing on the test rows in a couple of months.

Elon Musk (46:36):

Yeah. Although the line won't be obviously, at this rate-

Lars (46:40):

Initially.

Elon Musk (46:41):

… initially, this is a revolutionary production system. I'm not sure what the right word is. Unboxing sounds like something… Like when you get your phone.

Lars (46:52):

You open it up.

Elon Musk (46:53):

Yeah, you have a pleasant experience when you take your phone out of the box, which, of course, is nice, but this is much more revolutionary than that. This is a profound re-imagining of how to make cars in the first place. No car is made like this anywhere in the world. The factory is the product as much as the car is the product. It's really just the first principles approach to manufacturing that will ultimately allow us, I think, to… I'm trying to think. I'm confident, ultimately, it'll allow us to achieve a cycle time, meaning a unit every five seconds or less, off a single line.

Speaker 2 (47:43):

And we want to incorporate some of these for testing into our existing production lines as well to [inaudible 00:47:49] Cybertruck already.

Elon Musk (47:51):

This is something I've been thinking about for a long time and I've been thinking about this for a long time. It's not a crazy thing. A car every five seconds may sound like it's coming out like bullets, but actually it's coming out at walking speed.

Speaker 2 (48:08):

It's like a meter a second.

Elon Musk (48:09):

A meter a second. We're still far away from caring about the aerodynamic drag of the manufacturing line because you're still at three miles an hour type of thing. Every five seconds sounds crazy, but it's three miles an hour we're talking about. Yeah, you can run away from it basically. But that's still by far the fastest line on earth. It's half… I don't know. What's the [inaudible 00:48:51] line? I don't know if it's about-

Speaker 2 (48:52):

Shanghai phase two.

Elon Musk (48:53):

And that's us?

Speaker 2 (48:54):

33 seconds. Yes.

Elon Musk (48:55):

We're the fastest, right?

Speaker 2 (48:56):

I would think so.

Elon Musk (48:58):

We think we're the fastest at 33 seconds in our Shanghai factory, but this would be six times faster or seven times faster, more advanced. It'll be slower than that at first. But the point is that, when you fully optimize the design, an operation of the next generation factory that we're building right now, a five-second cycle time or less is… The design is capable of it.

(49:39)
When you go through a radical new architecture, you go from being in A… I'd say Shanghai, in particular, is an A+ on a moderately… An advanced, but still…

Speaker 2 (49:53):

Traditional?

Elon Musk (49:54):

Traditional car production system. They're really doing about as good as possible to do within a conventional scenario. Trying to get much below 30 seconds is extremely difficult, but if not… And you start getting into impossible… Where you have to be faster than a human could possibly move. Then the autonomous line, it really just needs to be robots moving really fast and that's where it gets us up five seconds. But we'll start off with getting a C… Instead of an A getting a C in a new architecture, but then the potential is there over time to move that up to an A+ within an A+ architecture.

Travis Axelrod (50:49):

Great. Thank you very much. The next question is how is Tesla positioning itself to flexibly adapt to global economic risks in the form of tariffs, political biases, et cetera.

Speaker 3 (50:59):

As Elon said, we've been a supply chain team for a while. We continue to mitigate global economic risks like tariffs and political biases by regionalizing parts, supplying near its factories in North America, Berlin and Shanghai. For example, in North America, our high volume vehicle programs have over 85% North American content and Shanghai vehicles have over 95% local content on similar levels of regionalization as North America when you exclude the battery. And we are working on regionalizing the battery as well.

(51:34)
This is a pre-pandemic strategy that we accelerated post-pandemic through supply diversification, tool sourcing, vertical integration, advanced analytics and local partnerships to ensure supply chain resilience and production stability. Having said that, we are not 100% insulated and these tariffs are higher on our low volume platforms than the high volume ones.

Elon Musk (51:58):

Yeah. In fact, there's no more vertically integrated car company than Tesla. We're the most vertically integrated car company since Henry Ford back in the day, when they were mining iron and stuff and growing rubber trees. We're not growing rubber trees and mining iron yet, but we have built a lithium refinery in South Texas and it's, I think, the biggest lithium refinery outside of China, I think. Is that right?

Speaker 3 (52:35):

Yeah.

Speaker 4 (52:35):

I think so.

Speaker 3 (52:35):

[inaudible 00:52:38].

Elon Musk (52:39):

But it's output potential would be the biggest lithium refinery outside… And we've got space to expand it if we need to build more, right? And then we've got the cathode refinery in Austin next to the Gigafactory. We've got to figure out what to do about the anode. That's an ongoing subject of discussion. The best of all possible worlds would be figuring out how to have no anode. Best part being no part. That's the dream of the lithium batteries to the anode. Not have an anode. But either way, we better have the anode, the cathode, and the lithium and the electrolytes and the separator to make a cell. But there's no other car company that is built in lithium refineries and cathode refineries. We're ridiculously vertically integrated and that's our best position to protect against supply chain disruptions. [inaudible 00:53:45], do you want to talk about progress in the front?

Speaker 4 (53:47):

Yeah. Certainly for our in-house cells, we've multi-sourced every component and we have every part coming from at least two different countries of origin, which is… We started this… The supply chain team and the engineering team worked together on this for the last couple of years to put that together. It's not something we did in a couple of months. This is years of work. We're in a good position to take advantage of that. And the insourcing of lithium and cathode, they're the two most critical parts of the battery that's run in that backyard. And we've total insulated from…

Elon Musk (54:26):

Oil. It needs to be an operation.

Speaker 4 (54:29):

It needs to be an operation. That's the right way to-

Elon Musk (54:31):

We also make our own cells, by the way. Cell production… If you make the anode, the cathode, the lithium, the electrolyte separator, the can, and then you got to put all that together in the cell factory. And there are entire companies that all they do is produce cells, but they don't do the other stuff. They don't refine lithium or the cathode. Our cell production is going quite well and I think we're… We're currently the lowest cost per kilowatt-hour

Speaker 4 (55:10):

Yeah. For all cells we purchased in North America.

Elon Musk (55:15):

Yeah.

Speaker 4 (55:15):

The lowest cost to us, I think.

Elon Musk (55:18):

We have the lowest cost per kilowatt-hour, all things considered. The Tesla cell is the most competitive sell for a kilowatt-hour put into a car. If it's a Tesla cell, it's lower cost than if it's a supplier sell.

Speaker 4 (55:32):

And the plan this year is to really build off that base. Getting to lowest cost is… It's the hardest challenge for so many batteries. It's relatively easy to build a flashy product that does one thing well, to build something at high volume and low cost is super difficult and we're using that as base to then build off and add performance in different areas for new products coming out.

Elon Musk (55:59):

Yeah.

Karan (56:05):

Yeah. To Elon's point, there's a lot of advantages for regionalization. The most important thing is we're not tying the working capital for six to eight weeks on the ocean. If there's a design change, then everything that's in transit basically has to be scrapped. Secondly, poor disruptions, as we saw during COVID, can be very expensive because slight disconnects can shut down production, so then your only option is costly air expedite.

(56:27)
It also gives us resilience and supply chain. If one region is down, we can bridge with others. It's more to set up in the beginning, but it's critical to have when the need arises. Having said that, it's unrealistic to zero on 100% regionalization across the board for specialized areas such as semiconductors. In such cases, our team works very closely with our partners to ensure we have strategic banks in place and a disruption doesn't impact production while we stand up the regional manufacturing for that particular commodity.

Lars (56:54):

And I'll say, on the rest of the vehicle, like Elon was talking about with cells, we're also heavily for vertically integrated, import ingots, internment castings, we recycle those in melt centers. The same thing with plastics, but it doesn't mean we're not exposed. We do have some areas where we use rare earth magnets. We've been working for years to find alternative sources and bring those up as well as we have our induction machines. And as we've mentioned in the past, we're working our fairing numbers for some time. As Karan said, with our heavy regionalization percentages, we're definitely the lowest exposed to this, but we're not completely immune, as [inaudible 00:57:33] mentioned in his opening remarks.

Travis Axelrod (57:35):

Great. Is it similarly related on the battery guide? Does Tesla still have a battery supply constraint, as noted on the Q4 call and does that change with tariffs?

Karan (57:47):

This is Karan. We've been working very hard to expand battery cell production in the US, both with vendors and, what Bonnie mentioned earlier, with the 4680 program. And we're also working on moving the upstream supply chain for battery cells to the United States for several years. And that strategy is really starting to pay off now. As it stands right now, we're not constrained on battery self supply for vehicles. The recent tariffs do pose some challenges to Tesla energy, like our CFO mentioned earlier, but it's something we've been anticipating and we should be able to resolve in a timely fashion. We actually have a plan to find a place where we're executing towards it.

(58:23)
We also have some other sources coming online to supplement the shortfall. And then, of course, we have the LL3 production that's happening in-house. We have a slight disconnect of aligning the right cells with the right pack, so that's the little bit of puzzle that we have to solve internally. But as far as cells go, there's no shortage.

Travis Axelrod (58:45):

Great. Thank you very much. The next question is, did Tesla experience any meaningful changes in order inflow rate in Q1 relating to all of the rumors of brand damage?

Speaker 2 (58:58):

In Q1, as mentioned earlier, we took the best-selling car over the last two years and ramped up all four of our global factories. And in less than eight weeks, we've already gone to the rate of our previous Model Ys, the factories. Just kudos again to the team for the great job there. And despite the economic strain and negative articles, in California in Q1, tesla remained the best-selling car, not just EV. And additionally, we had a record number of test drives globally in Q1 as well, so interest remains high. And so, right now we continue to see good interest still on the vehicle.

Elon Musk (59:39):

Yeah, Tesla is not immune to the macro demand for cars. When there is economic uncertainty, people generally want to pause on doing a major capital purchase like a car. But as far as absent macro issues, we don't see any reduction in demand.

Speaker 2 (01:00:10):

Correct. That's where we continue to focus on affordability and it's a fundamental focus there.

Travis Axelrod (01:00:17):

Yeah. Fantastic. Thank you, guys. The next question is regarding the Tesla Optimus pilot line. Could you confirm if it is currently operational? If so, what is the current production rate of Optimus bots per week? Additionally, how might the recent tariffs impact the scalability of this production line moving forward?

Elon Musk (01:00:40):

I just want to emphasize, Optimus is still very much a development program. It's not a large volume production. That's why this year we will make a few. We do expect to make thousands of Optimus robots, but most of that production is going be at the end of the year. Almost everything in Optimus is new. There's not an existing supply chain for the motors, gearboxes, electronics, actuators, really almost anything in the Optimus, apart from the Tesla AI computer, which is the same as one in the car.

(01:01:26)
When you have a new complex manufactured product, it'll move as fast as the slowest and least lucky component in the entire thing. And as a first-order approximation, there's 10,000 unique things. That's why anyone here who tells you they can predict, with precision, the production ramp of a truly new product doesn't know what they're talking about. It is literally impossible. You go through this series of constraints where this part's a limiting factor, now that part's a limiting factor, now this part's a limiting factor, and multiply that by 1000, basically. And then the rate of the production ramp is decided by how quickly you can solve each of those problems. Now, Optimus was affected by the magnet issue from China because the Optimus actuators in the arm do use permanent magnets.

(01:02:36)
Now, Tesla as a whole does not need to use permanent magnets, but when something is volume constrained, like an arm of the robot, then you want to try to make the motors as small as possible, so we just did the design in permanent magnets for those motors, and those were affected by the supply chain, by basically China requiring an export license to send out anywhere with magnets.

(01:03:09)
We're working through that with China. Hopefully we'll get a license to use the rare earth magnets. China wants some assurances that these are not used for military purposes, which obviously they're not, they're just going into a humanoid robot. It's not a weapon system, but that is an example of a real challenge there. I'm confident we'll overcome these issues and we'll, by the end of this year, have thousands of Optimus robots.

Travis Axelrod (01:03:40):

Great. Thank you very much. And the last question we already covered earlier, whether Robotaxi was still on track for this year. With that, we can move on to analyst questions. The first question is going to come from Pierre at New Street. Pierre, please unmute yourself.

Pierre (01:04:02):

Hey, guys. Can you hear me?

Elon Musk (01:04:04):

Yeah.

Pierre (01:04:05):

That's great. I'm super excited to hear Robotaxi and Optimus becoming the very tangible future for Tesla. But I have actually a question on the legacy… Not the legacy, but the current auto business. And when I look back at the ramp of Model 3 a few years ago, I really saw it as being the iPhone of cars. A new product completely reinvented, very different user experience, vastly superior, impossible to match for traditional competitors. And for the iPhone, it resulted in the high end of the smartphone market quadrupling in size and, actually, Apple taking 60% market share.

(01:04:49)
And so, when you look at the Model 3 in the Model Y today, I think they are still really vastly superior to any other cars. And I wonder why they've taken about 15% of their addressable market and not more, actually. Another way to put it is why are there so many people still buying BMWs and Mercedes, knowing that the Model 3 and the Model Ys are out there and available? And I wonder if you're trying to solve that [inaudible 01:05:19] internally, if you understand why, what are these auto buyers who are not buying a Model 3 or Model Y missing? And if you have ideas of things you could do to address that. Maybe there is enormous value left on the table there. That's what I'm wondering these days.

Elon Musk (01:05:40):

Yeah. The reality is that, in the future, most people are not going to buy cars. One could say, look, if you want to continue with your phone metaphor, you can remember the days of the flip phones, when there was

Elon Musk (01:06:00):

A hundred different flip phone designs. And the mistake that manufacturers made was to try to make many different variants of a flip phone, which was a mistake. They should have made the iPhone. So because obviously everyone's going to want a smartphone. But in the beginning when the iPhone came out, I was like, "Wow, I can't believe these guys are not reacting as though this is death." But they didn't, they kept making different variants of flip phones. Nokia, I think at one point was the most valuable company in the world, or close to it, but they kept making flip phones, trying to find another market niche. Maybe somebody wants a phone of a different style, maybe this different color, or whatever it is. Nope, they just want a super intelligent phone that can do everything. Just one. I said this many years ago. In the future, in the not too distant future, buying a gasoline car that is not autonomous will be like riding a horse while using a flip phone. Some people still do it, but it's rare.

Speaker 5 (01:07:39):

Great. Thank you. The next question comes from Emmanuel Rosner at Wolfe. Emmanuel, please unmute yourself.

Emmanuel Rosner (01:07:50):

Great. Thanks for taking my question. So Elon, the public version of the FSD software still has a decent amount of, I guess, intermittent human interventions that are required. So what's still required for the software on your end to get to a level where it doesn't need to be supervised? And I'm asking that in the context of obviously the June launch being in the next couple of months. What still needs to happen?

Joshua (01:08:23):

We are working on a number of items too.

Elon Musk (01:08:27):

Yeah, go ahead Joshua.

Joshua (01:08:27):

We are aware of the interventions that are happening in the public biz, and that's why we are hardcore burning it down. And really speaking, some initial launch [inaudible 01:08:39] helped us focus on solving all the issues that you face here. For example, I guess, focusing on Austin, we are not solving all the issues that customers in Boston or somewhere else might face. And then here we just have big list of all the issues, just burn it down, and that's what the team is working on, along with other sort of redundancy issues. For example, if one month the computers goes down, right, and the customer fleet, it would throw the red hands and ask you to take over, but we don't want that kind of situation. So you're solving both the reliability issues of the autonomy software and also the reliability issues of the system software together for Austin.

Elon Musk (01:09:21):

Yeah, really we just work through a long tail of unusual interventions. And these are really very rare. As I was saying earlier, you can have an intervention every 10,000 miles. I mean, that's a lot of driving you've got to do to even find one case within Athens.

Joshua (01:09:42):

Yeah, and some interventions…

Emmanuel Rosner (01:09:42):

[inaudible 01:09:43].

Joshua (01:09:43):

… due to systematic missing functionality. For example, for handling emergency vehicles correctly, you don't need to consume audio as an input. But then the customer facing versions don't have audio input, but the version that's going to be in Austin, will have audio input, and so on.

Emmanuel Rosner (01:10:03):

Okay, but would you have remote operators, for example?

Joshua (01:10:08):

Every now and then if a car gets stuck or something, someone will unblock it, but it's just because we are a bit conservative and tend towards more safety than even if we get stuck every now and then. We do have remote support, but it's not going to be required for safe operation. If anything, it's just required for more availability.

Elon Musk (01:10:30):

Anyway, it's only a couple months away, so you can just see for yourself in a couple months in Austin.

Speaker 5 (01:10:37):

Great. Our next question comes from Edison at Deutsche Bank. Edison, please unmute yourself.

Edison (01:10:48):

Hi. Thank you very much for the question. So I wanted to ask about the optimist supply chain, going forward. You mentioned very fast ramp up. What do you envision that supply chain looking like? Is it going to require many more suppliers to be in the US now because of the tariffs? How does one think about what needs to happen there?

Elon Musk (01:11:12):

We have to see how things settle out. I don't know yet. Some things we're doing as we already talked about, which is that we've already taken tremendous steps to localize our supply chain. We're more localized than any other manufacturer, and we have a lot of things underway to increase the localization, to reduce supply chain risk associated with geopolitical uncertainty. Did you have followup, Edison?

Edison (01:11:50):

Yeah. Wanted to come back actually to the RoboTaxi, then. Do you have a sense on how many cars or how big the scale will be initially, and how that might ramp up? I know you're targeting millions of vehicles in the second half of next year, but initially at launch, how many vehicles would be reasonable, and is it going to be as simple as if one goes to Austin, let's say in late June or July, you'll be able to request?

Elon Musk (01:12:18):

Yeah. We're still debating the exact number to start off on day one, but it's like, I don't know, maybe 10 or 20 vehicles of day one. And watch it carefully, then scale it up rapidly after that. So we want to make sure that we're paying very close attention the first time this happens. But yeah, you'll be able to, end of June or July, just go to Austin and order a Tesla Truck, autonomous drive.

Speaker 5 (01:12:53):

Great. The next question comes from George at Canaccord.

George (01:13:00):

Hi. Thank you for taking my question. It has to do with FSD pricing. Could we envision when you launch unsupervised FSD that there could be sort of a multi-tiered pricing approach to unsupervised versus supervised, similar to what you did with Autopilot versus FSD in the past? Thank you.

Vaibhav Taneja (01:13:19):

I mean, this is something which we've been thinking about, I mean, just so you know, for people who have been trying FSD and who've been using FSD, they think even the current pricing is too cheap. Because for 99 bucks, you're basically getting a personal show.

Elon Musk (01:13:41):

Yeah, I mean we do need to give people more time to… A key break point is can you read your text messages or not?

Vaibhav Taneja (01:13:51):

Yes.

Elon Musk (01:13:52):

Can you write a text message or not? Because obviously people are doing this, by the way, with unautonomous drives all the time. And if you just go for a drive down the highway and you'll see people texting while driving, doing 80 mph.

Lars (01:14:05):

And putting makeup on at the same time.

Elon Musk (01:14:06):

Yeah. Putting on makeup, doing their hair, with the mirror down, and texting, and driving, at 80 mph. This is a common occurrence. So people eating lunch, you name it, shaving. So anyway, but right now the car is very insistent that you pay attention to the road, which reduces the value somewhat, because it's very rigorous about you paying attention to the road. And we will gradually lighten up on that, with every few weeks or every month, we'll relax that a little bit, so you can be more and more able to do things you want to do and not have the car demand your [inaudible 01:15:05] attention. So that value, it'll really be profound when you can basically do whatever you want, including sleep. And then that $99 is going to seem like the best $99 you've ever spent in your life.

Speaker 5 (01:15:24):

Great. And George, did you have a follow up?

George (01:15:27):

My follow up is about geographic expansion. Just maybe discuss additional markets. There's been some news around India recently that you could launch this year and next. Thank you.

Vaibhav Taneja (01:15:38):

So yeah, we've been working on getting into India. India is a very hot market. And especially the current, and I don't want to talk just about tariffs, but the current structure with India is that any car which we send in is subject to 70% tariff. Also on a 30% luxury tax on it. So the same car which we're selling is 100% more expensive than what it is. So that creates a lot of anxiety, is people feel okay, they're paying too much for the car. And by the way, we are not getting the money, the local government is getting the money. And that's why we've been very careful trying to figure out when is the right time. Like I said, we are working on it, it would be a great market to enter, because India has a big middle class, which we would want to tap in, and that is the market which we want to be in. But again, these kind of things create a little bit of tension, which we're trying to work around.

Speaker 5 (01:16:46):

Great. Thank you so much. The next question comes from Adam Jonas at Morgan Stanley. Go ahead, Adam. We can't hear you Adam. So maybe we'll put you back in the queue and we'll move to Colin Langan from Wells Fargo, while Adam figures out this audio. Colin, please unmute yourself.

Colin Langan (01:17:21):

Oh, great. Do you hear me?

Speaker 5 (01:17:23):

Yes.

Colin Langan (01:17:24):

Oh, great. You're still sticking with the vision only approach. A lot of autonomous people still have a lot of concerns about sun glare, fog, and dust. Any color on how you anticipate on getting around those issues? Because my understanding, it kind of blinds the camera when you get glare and stuff.

Elon Musk (01:17:44):

Actually it does not blind the camera. We use an approach which is direct photon count. So when you see a processed image, so the image that goes from the silicon photon counter, that then goes through a digital signal processor or image signal processor. That's normally what happens. And then the image that you see looks all washed out because if you point the camera at the sun, the post-processing of the photon counting washes things out, it actually adds noise. So quite a big breakthrough that we made some time ago was to go with direct photon counting and bypass the image signal processor. And then you can drive pretty much straight at the sun, and you can also see in what appears to be the blackest of night. And then here in fog, we can see as well as people can, probably better. But in fact, I'd say probably slightly better than people, than the average person, anyway. And, yeah.

Colin Langan (01:19:11):

So the camera is able to see when there's direct glare on it?

Elon Musk (01:19:14):

Yeah.

Colin Langan (01:19:14):

I'm a little surprised by that.

Elon Musk (01:19:17):

Yeah.

Colin Langan (01:19:18):

Okay. And then just there were obviously media reports the other day that the affordable model was delayed, doesn't sound like That's correct. Those reports also talked about it being more of a cheaper version of the model Y. any color on what we should expect? Is it a cheaper version of the model Y or is it actually going to be a design change with it?

Vaibhav Taneja (01:19:41):

So I think Lars only covered it in answering one of the say.com questions. The real thing which we are trying to focus on is affordability, and using our existing lines, and there's always limitations when you're using existing lines as to how many different form factors can you bring through. So that's the way I would say you should think about it. And I don't know if Lars anything more to add.

Lars (01:20:07):

Yeah, I think I said this before in other calls. With the recent upgrades to the Model 3 and the Model Y platforms, we made some pretty great cars at pretty great prices, and added a bunch of features and things like that. I think it's easy to consider that moving forward, Tesla doesn't make bad cars, and our intent is not to make a car that is any worse than any car we've ever produced in the past. And so the models that come out in the next months will be built on our lines and will resemble in form and shape, the cars we currently make. And the key is that they'll be affordable and you'll be able to buy one.

Speaker 5 (01:20:41):

Great. We might have time for one last question. Adam, we'll try your audio again. You want to try to unmute yourself, Adam. All right. Unfortunately still not working.

Adam Jonas (01:21:00):

Yeah. Hi.

Speaker 5 (01:21:02):

Oh. There you go.

Adam Jonas (01:21:04):

Oh. Sorry guys. Technology.

Speaker 5 (01:21:08):

Go ahead, Adam.

Adam Jonas (01:21:09):

Yeah, hi. Yeah, in the February 28th, Joe Rogan interview, Elon, you advocated for a ramp in tariffs to give people time to adjust, otherwise, I quote, you said, "The system would break and bad things would happen." So are things breaking yet? And if the tariffs as announced remain in place, when would things start breaking?

Elon Musk (01:21:33):

Well, at the risk of stating the obvious, I'm one of many advisors to the president, I'm not the president. But I've made my opinion clear to the president, and other people made their opinion clear to the president. He talks to many people and he makes his decision. And I'm hopeful that the President will observe whether my predictions are more accurate than the predictions of others, and perhaps weigh my advice differently in the future, which we'll see. But I'm an advocate of predictable tariff structures, and generally I'm an advocate for pre-trade and lower tariffs. Now one does need to take a look at if some country is doing something predatory with tariffs or if a government is providing extreme financial support for a particular industry, then you have to do something to counteract that. But I think that that's on a case by case basis, strategically. But the president is the elected representative of the people and is fully within his rights to do what he would like to do.

Adam Jonas (01:23:09):

Okay, Elon, I respect that. Just as a follow-up, and thanks again, between China and the United States, who in your opinion is further ahead on the development of physical AI, specifically on humanoids? And also drones, I'd be interested. And is it even close? Yeah, I'm serious, and [inaudible 01:23:46].

Elon Musk (01:23:46):

Well, I think you know the answer on the drones. I mean, a friend of mine, Naval, posted on X, I reposted it, I think a prophetic statement, which is, "Any country that cannot manufacture its own drones is doomed to be the vassal state of any country that can."

And America cannot currently manufacture its own drones. But that's [inaudible 01:24:10] fortunately. So China I believe manufactures about 70% of all drones. And if you look at the total supply chain, almost 100% of drones have a supply chain dependency on China. So China is in a very strong position, and here in America, and we need to shift more of our people and resources to manufacturing, because this is…

(01:24:43)
And I have a lot of respect for China, because I think China is amazing actually, but the United States should not have such a severe dependency on China for drones and be unable to make them, unless China gives us the parts, which is currently the situation. With respect to humanoid robots, I don't think there's any company in any country that can match Tesla. Tesla and SpaceX are number one. Now I am a little concerned that on the leaderboard, ranks two through 10 will be Chinese companies, but I'm confident that rank one will be Tesla.

Speaker 5 (01:25:46):

Great. Well, I think that's unfortunately all the time we have for today. We appreciate all your questions and look forward to talking to you next quarter. Thank you very much and goodbye.